Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C031 **IDP1** 4 genes / 2 species 3 direct pairs	**Y1 IDP1** GenAge 1346 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 18340043, 20421943 **Y2 IDP2** GenAge 1366 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043, 20883526 **Y3 IDP3** GenAge 1642 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 18340043, 19030232	**W1 idh-1** GenAge 275 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 15998808	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (n:m); W1–Y2 (n:m); W1–Y3 (n:m)				
C090 **INP53** 2 genes / 2 species 1 direct pair	**Y1 INP53** GenAge 1367 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 16293764, 18340043 20883526	**W1 unc-26** GenAge 703 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 9789046	No gene in this candidate network	No gene in this candidate network
Direct pairs: W1–Y1 (1:m)				
C106 **ire-1** 2 genes / 2 species 1 direct pair	No gene in this candidate network	**W1 ire-1** GenAge 948 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 20460307	**F1 Ire1** GenAge 2148 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 27783936	No gene in this candidate network
Direct pairs: W1–F1 (1:1)				

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.